Aqyl Jol: An AI-Based System for Urban Traffic Management and Municipal Fleet Coordination

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***Abstract*—Rapid motorization in growing cities exacerbates congestion, delays municipal road maintenance, and degrades urban mobility. This paper presents *Aqyl Jol*, an integrated AI-based pipeline for municipal hazard-response support. The technical core couples computer-vision-based traffic and road-condition perception with priority-aware fleet coordination. The perception layer combines a YOLOv8 detector for traffic participants and three YOLOv11 road-condition segmentation models for flood, snow, and ice analysis from city cameras. These outputs are fused by an Intelligent Resource and Route Optimization module (IRKT) that ranks road segments and recommends suitable vehicle dispatch with route constraints. A FastAPI microservice backend and an iOS client expose map-based monitoring, incident reporting, and multilingual conversational access through an LLM interface layer. Reported experimental results provide component-level and integration-level evidence of operational feasibility for a unified smart-city platform.**

***Index Terms*—smart city, intelligent transportation systems, computer vision, large language models, municipal fleet management, mobile application**

I. Introduction

Rapid urbanization and the growth of private motorization have led to persistent congestion, increased emissions, and rising operational pressure on municipal service fleets. Vision-based intelligent transportation systems (ITS) have shown strong utility for traffic monitoring, density estimation, and incident-aware control [1]–[4]. In parallel, smart-city platforms increasingly incorporate natural-language interfaces to improve access to traffic and municipal information [5]–[7].

Despite this progress, many deployments remain fragmented: perception is often evaluated separately from operational response, while user-facing interfaces are treated as stand-alone services. For municipal road hazards (e.g., snow, flooding, ice), this separation limits reproducible end-to-end workflows from detection to dispatch support.

To address this gap, we study *Aqyl Jol* as an integrated municipal hazard-response pipeline that links camera-based perception, priority-aware dispatch support, and operator/citizen access layers.

The main contributions are:

- An integrated operational pipeline that links visual road-state perception with municipal response support through a shared analytics and service stack.
- A modular computer-vision workflow for traffic and road hazards using a YOLOv8 detector and three YOLOv11 segmentation models on Astana camera data.
- A deployment-oriented smart-city service layer that exposes structured outputs via map overlays, routing constraints, and multilingual natural-language access.

II. Related Work

Vision-based traffic monitoring has matured substantially, especially for density estimation, vehicle detection, and flow analysis in urban scenes [1], [3], [4]. Prior approaches typically optimize perception accuracy and runtime behavior at the model level, including practical deployment on traffic-monitoring infrastructure [2]. In contrast, the present work uses perception as one stage within a broader municipal-response pipeline.

For municipal operations, road-condition perception (including snow, water, and ice cues) is increasingly relevant because response quality depends on timely and spatially localized hazard evidence [8], [9]. Existing studies focus primarily on hazard recognition quality, whereas downstream coupling to prioritization and fleet-facing decision support is usually not explicit.

Operational dispatch and routing modules in ITS are often studied independently from perception pipelines [10], [11]. As a result, perception outputs and dispatch logic are frequently connected only loosely at system level, which limits reproducibility of end-to-end municipal workflows under practical constraints.

LLM-based interfaces for smart cities are a recent line of work [5]–[7]. Prior approaches typically position conversational models as access layers for information services; similarly, this paper treats the LLM as a multilingual interaction layer rather than a core optimization engine.

Taken together, the main gap in the literature is not a single missing detection model, but a missing integration pattern across perception, prioritization, and service delivery. This paper addresses that gap by connecting camera-based traffic/hazard perception, IRKT-based segment prioritization, and user/operator-facing service layers within one operational pipeline.

III. System Overview

We consider an urban road network where segments $r \in \mathcal{R}$ are monitored over discrete time intervals t. Municipal fleets—sand/salt spreaders, plow trucks, water trucks, tow trucks—operate over this network. The architecture comprises four layers: a *data layer* ingesting video streams, fleet telematics via the Omnicomm platform, and mobile interaction logs; an *analytics layer* running CV models and the IRKT prioritization engine; a *service layer* exposing RESTful APIs via FastAPI microservices in Docker containers on AWS; and an *application layer* comprising an iOS app built on the 2GIS SDK for mapping, incident reporting, crew tracking, and conversational access.

IV. Methodology

A. Data Sources and Preprocessing

1) Traffic Video and City Camera Data: We use a CV pipeline based on YOLOv8-S, fine-tuned on urban traffic imagery from Astana comprising 14,026 training images, 1,334 for validation, and 667 for testing across 23 object classes (cars, trucks, buses, bicycles, pedestrians, traffic lights, etc.). Images are resized to 512×512 pixels with rectangular batching. Aggressive augmentations (mosaic, mixup) are disabled; mild blur and grayscale perturbations are retained to match roadside camera statistics. The dataset exhibits a long-tail distribution where frequent vehicle classes dominate. Additionally, pixel-level road-condition segmentation masks from three specialized datasets (flood, snow, ice) support hazard detection and congestion analysis.

2) Municipal Fleet and Operational Data: Fleet data include GPS trajectories, vehicle types, and operational events from the Omnicomm telematics platform. Due to limited local access, we supplemented with publicly available datasets matched to local vehicle categories. Each record is mapped to segment r and interval t, enriched with road class, free-flow speed, and weather conditions.

3) Mobile Application Logs: The iOS application collects navigation requests, LLM chat interactions, and incident reports stored in Firebase Firestore with per-user anonymized documents.

B. Computer Vision Pipeline

1) Object Detection and Classification: The YOLOv8-S detector uses a ResNet18-based backbone with C2f blocks, SPPF layers, and decoupled classification/regression heads. The training loss is:

$$L = \lambda_{\mathrm{box}} L_{\mathrm{CIoU}} + \lambda_{\mathrm{cls}} L_{\mathrm{CE}} + \lambda_{\mathrm{dfl}} L_{\mathrm{DFL}}, \quad (1)$$

where L_{CIoU} is the Complete-IoU loss, L_{CE} is classification cross-entropy, and L_{DFL} is Distribution Focal Loss.

Training was conducted on an NVIDIA L4 GPU (Python 3.12, PyTorch 2.8.0 with CUDA) with batch size 96, up to 25 epochs with early stopping, and mixed precision. On the test set, the detector achieved AP@0.5 of 0.696 for cars, 0.575 for trucks, and 0.614 for bicycles, with aggregate mAP@0.5:0.95 $\approx$ 0.134. Inference latency is 8–16 ms per image (up to 125 FPS). Fig. 1 shows sample detection outputs on Astana imagery.

2) Road Condition Segmentation: Three YOLOv11-based semantic segmentation models are trained:

$$L = \lambda_{\mathrm{box}} L_{\mathrm{CIoU}} + \lambda_{\mathrm{cls}} L_{\mathrm{CE}} + \lambda_{\mathrm{dfl}} L_{\mathrm{DFL}} + \lambda_{\mathrm{mask}} L_{\mathrm{mask}}, \quad (2)$$

where L_{mask} is the mask loss computed using Binary Cross-Entropy (BCE) over pixel-wise predictions, and the other terms remain consistent with the detection formulation.

Each frame produces a per-pixel class map $M_t(x, y) \in \{\mathrm{road}, \mathrm{snow}, \mathrm{water}, \mathrm{ice}, \mathrm{other}\}$, used for safety alerts and as inputs to the IRKT dispatch engine.

3) Traffic Density and Congestion Estimation: Detections are filtered to vehicle classes on drivable areas using the segmentation mask. For road segment r, density $\rho_r(t) = N_r(t)/|\Omega_r|$ is computed with class-specific weighting. Flow is estimated via virtual counting lines and multi-frame tracking. A normalized congestion index $C_r(t) \in [0, 1]$ categorizes traffic states. Validation confirms accuracy of 78–85%.

- **AstanaFlood:** 480 images (flooded buildings, roads, water, vehicles); mAP@0.5 ≈ 0.694.
- **AstanaSnow:** 2,152 images (snow road, clear road, object); mAP@0.5 ≈ 0.852 for snow, 0.825 for clear road.
- **AstanaIce:** 785 images with pixel-wise ice masks; mAP@0.5 ≈ 0.715.

C. Intelligent Resource and Route Optimization (IRKT)

The IRKT module is implemented as a heuristic priority-ranking and vehicle-assignment procedure driven by perception outputs and fleet telemetry. For each road segment r at time t, IRKT combines: (i) hazard severity from the CV hazard map (ice/snow/flood), (ii) traffic exposure from density/congestion estimates, (iii) road criticality (e.g., class and proximity to critical infrastructure), and (iv) operational urgency.

Fig. 1. YOLOv8-S detection results on Astana city imagery. Left: congested traffic with bus and car detections (confidence 0.48–0.86). Center: mixed scene with trucks, cars, and pedestrians. Right: intersection with buses (0.88–0.89) and cars at varying distances.

This can be expressed as a policy-weighted score

$$P_r(t) = \alpha H_r(t) + \beta C_r(t) + \gamma K_r + \delta U_r(t), \tag{3}$$

where $H_r(t)$ denotes hazard severity, $C_r(t)$ denotes traffic exposure, K_r denotes segment criticality, and $U_r(t)$ denotes urgency terms derived from operational context. The coefficients are policy-configured heuristic weights rather than learned parameters.

Segments are ranked by $P_r(t)$, and assignment is computed over currently available compatible vehicles, selecting the nearest feasible vehicle for each high-priority segment. The output is a dispatch recommendation set containing ranked segments, selected vehicles, and route/ETA estimates. As new events arrive, the ranking and assignments are updated, and active treatment areas are propagated to routing as excluded zones.

D. LLM-based Assistant and Backend Services

The conversational component (`llama_service`) serves primarily as a multilingual interaction layer between users, operators, and the platform's analytical modules. It is powered by a locally deployed LLaMA-family model exposed through an OpenAI-compatible REST API and supports Russian, Kazakh, and English.

For user-facing queries, the assistant provides conversational access to traffic conditions, hazard alerts, route restrictions, and municipal service information. Responses are grounded in structured backend data, including congestion indices, fleet status, and active hazard zones. For operator-facing workflows, the assistant converts structured outputs from the platform into concise natural-language summaries, improving the interpretability of dispatch recommendations and detected incidents.

The query pipeline is as follows: (i) optional speech-to-text on the client; (ii) location extraction via 2GIS geocoding; (iii) retrieval of relevant traffic, routing, and operational context; (iv) LLM generation conditioned on the retrieved context; and (v) rendering in the mobile chat interface. This design allows the assistant to function as a human-centric access layer over the underlying perception and dispatch modules, rather than as an autonomous control component.

The backend comprises FastAPI-based microservices for computer vision, assistant inference, and routing/search, each containerized with Docker and deployed on AWS. The 2GIS SDK is used for geocoding, map rendering, and jam-aware routing with excluded areas corresponding to active hazard zones or maintenance operations.

E. Mobile Application

The iOS client (Swift 5, UIKit, iOS 15+) provides:

- A 2GIS map with real-time overlays for incidents, crew movements, congestion levels, and street closures.
- Bottom-sheet-driven UI (primary, search, info, route) with dynamic map padding.
- Routing for car, truck, and pedestrian modes with jam-aware search and excluded-area support via point, polygon, and polyline geometries.
- Chat interface with persistent Firestore-synced history, voice search via Apple Speech, and runtime-switchable localization (Russian, Kazakh, English) with dark/light themes.

The application serves as an interaction layer over system outputs rather than an independent analytical engine. User data are stored in Firebase Firestore with document-level access control, and communication uses HTTPS with modern TLS as part of privacy-aware data handling.

F. System Integration and Deployment

The system follows a service-oriented architecture that links sensing, analytics, dispatch support, and user/operator interaction. City cameras and fleet telemetry are ingested by the data layer; CV services then generate detections, road-condition masks, and congestion indices. These analytics feed IRKT, which produces segment priorities and vehicle-assignment recommendations.

At the service and application layers, outputs are delivered through 2GIS-based map overlays, routing with excluded areas, and multilingual conversational access. Backend components are deployed as containerized FastAPI microservices on AWS, supporting modular scaling for practical municipal deployment.

V. Experimental Evaluation and Results

A. Object Detection

The YOLOv8-S detector achieves mAP@0.5:0.95 $\approx$ 0.13 across 23 classes, with AP@0.5 of 0.60–0.70 for high-frequency vehicles. Inference runs at 60–125 FPS on an

NVIDIA L4. These values are consistent with real-time urban detection settings where latency constraints and long-tail class composition can depress aggregate mAP@0.5:0.95, while dominant operational classes retain substantially stronger AP. The observed pattern therefore reflects a practical operating regime: frequent vehicle classes, which drive congestion and dispatch decisions, are detected more reliably than rare classes.

As Fig. 1 shows, the model detects vehicles across diverse Astana scenes: dense traffic with buses and cars (confidence 0.48–0.86), intersections with trucks and pedestrians (0.35–0.87), and urban crossings with buses (0.88–0.89). Scores exceed 0.7 for clearly visible vehicles and remain above 0.4 for occluded or distant objects.

B. Road-Condition Segmentation

1) Flood Detection: Fig. 2 shows AstanaFlood metrics. The confusion matrix exhibits a strong diagonal with minor wet-surface/shallow-water confusion, indicating that dominant flooded-vs-nonflooded patterns are captured even when visually similar boundary regions remain ambiguous.

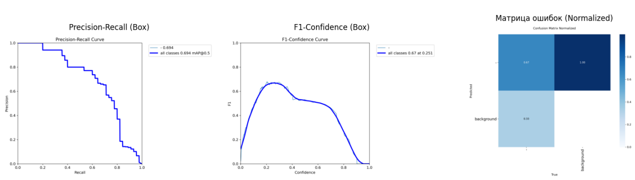

Fig. 2. AstanaFlood CV Output: Precision-Recall, F1-Confidence, and Normalized Confusion Matrix

2) Snow Analysis: AstanaSnow (Fig. 3) achieves AP of 0.852 for snow road and 0.825 for clear road. Aggregate mAP@0.5 of 0.563 is skewed by the sparse object class (0.012). This distribution is consistent with long-tail road-condition data, where dominant surface states are modeled more robustly than infrequent auxiliary classes.

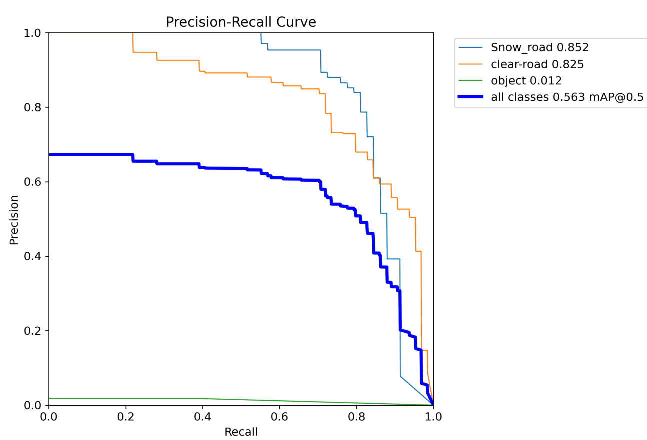

Fig. 3. AstanaSnow Precision-Recall Curve

3) Ice Detection: AstanaIce (Fig. 4) achieves mAP@0.5 of 0.709 with a high-precision plateau up to recall $\approx$ 0.6, enabling calibrated alerts. For municipal prioritization, this behavior is operationally useful because robust identification of high-risk ice regions is more critical than exact pixel-boundary recovery.

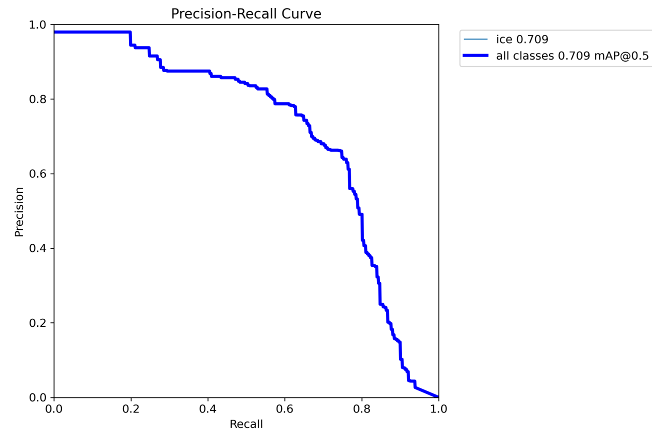

Fig. 4. AstanaIce Precision-Recall Curve

4) Overall Assessment: Across all three domains, the CV pipeline demonstrates sufficient fidelity to support hazard-aware routing and municipal fleet coordination. The AP levels for Snow road (0.852) and Ice (0.709) indicate that city surveillance imagery can provide useful real-time hazard signals for municipal operations. In this role, the approach can complement, and in some constrained scenarios partially substitute for, specialized road-weather sensing infrastructure.

The integration of detection and segmentation outputs with IRKT provides a closed operational loop: hazards are detected, segments are prioritized, crews are assigned, and treatment zones are exposed to users in near real time. While individual components are evaluated separately, their coupling enables a consistent perception-to-dispatch workflow, which is critical for practical municipal deployment where isolated perception metrics are insufficient.

At the same time, long-tail object classes and complex intersections remain important error sources, and current evidence should be interpreted as component-level and prototype-level feasibility rather than full municipal-impact validation.

C. Limitations

The present study has several important limitations. First, the evidence is based on component-level evaluation and integrated prototype behavior; it does not yet include a longitudinal municipal field deployment with measured network-level outcomes (e.g., sustained congestion reduction or response-time improvements). Second, IRKT is currently a heuristic policy-driven prioritization and assignment module rather than a learned optimization framework. Third, although the LLM layer improves multilingual accessibility and interpretability of structured outputs, it is not evaluated here as a primary standalone scientific contribution.

VI. Conclusion and Future Work

This paper presented Aqyl Jol as an integrated municipal hazard-response pipeline for urban traffic monitoring and fleet coordination. The main technical contribution is the coupling of camera-based traffic and road-condition perception with IRKT-based dispatch support, exposed through a unified service and application stack.

The reported results support operational feasibility at prototype level. The object detector runs in real time (60–125 FPS) with AP@0.5 of 0.58–0.70 for high-frequency vehicle classes, the road-condition models report mAP@0.5 between 0.69 and 0.85 for primary hazard classes, and congestion estimation reaches 78–85% accuracy against expert annotations. These findings indicate that integrated perception-to-dispatch workflows are technically viable for municipal use, while not yet establishing broad city-scale impact.

Future work includes: (i) integrating adaptive traffic signal control to complement fleet dispatch with network-level signal optimization; (ii) scaling the CV pipeline to additional cameras and contractor fleets with richer sensor fusion from roadside units; (iii) replacing heuristic IRKT scoring with a learning-to-route framework trained on historical dispatch outcomes; (iv) incorporating causal machine learning models to quantify deployment effects through counterfactual analysis; and (v) conducting longitudinal field studies with municipal partners.

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